Methods In Differential Equations

Solutions

Introduction

Differential equations relate variables and their derivatives. A first-order equation contains a first derivative but no higher derivatives.

Temperature loss proportional to temperature above the surroundings:

$$\frac{dT}{dt} = -kT.$$

Here T is excess temperature.

Population change proportional to $P(1 - P/M)$, with limiting population M (Verhulst–Pearl model):

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right).$$

GDP x grows at a rate proportional to its current value:

$$\frac{dx}{dt} = kx.$$

Applications in physics and engineering include radioactive decay, mixing fluids, cooling materials and bodies falling against resistance.

Pure Year 2 Recap

Solution 3

Separation of variables applies when the right-hand side is a product of one function of x and one of y:

$$\frac{dy}{dx} = f(x)g(y).$$

For $g(y) \neq 0$, divide through:

$$\frac{1}{g(y)}\frac{dy}{dx} = f(x).$$

Multiply by dx and integrate:

$$\int \frac{1}{g(y)}\,dy = \int f(x)\,dx.$$

Include an arbitrary integration constant. Check separately for constant solutions satisfying $g(y) = 0$.

Examples

Solution 4

Find the general solution of $dy/dx = 2$.

Solution $y = 2x + C.$

Why is it called the general solution?

Family

The arbitrary constant gives a family of solutions.

Find the general solution of $dy/dx = -x/y$, where $y \neq 0$.

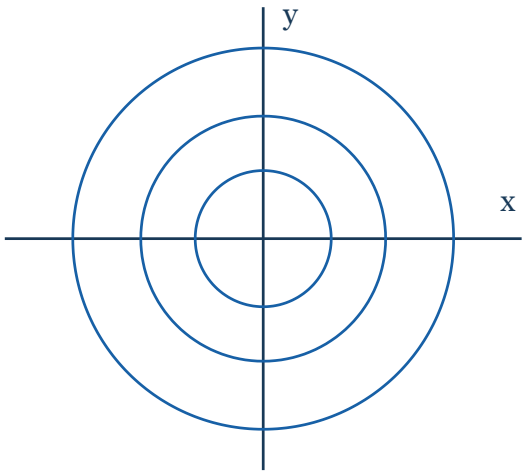
Solution

$$\int y\,dy = \int -x\,dx$$

$$\tfrac{1}{2}y^2 = -\tfrac{1}{2}x^2 + C$$

$$x^2 + y^2 = 2C = r^2.$$

The family of circles satisfies the equation wherever $y \neq 0$.



Examples

Solution 5

Find the general solution of $dy/dx = xy + x$.

Separating

$$\frac{dy}{dx} = x(y+1)$$

$$\int \frac{1}{y+1}\,dy = \int x\,dx$$

$$\ln|y+1| = \tfrac{1}{2}x^2 + C$$

$$|y+1| = e^{x^2/2+C} = e^{x^2/2}e^C$$

$$y + 1 = Ae^{x^2/2}.$$

Absorb the sign into real A. Including $A = 0$ restores the equilibrium solution $y = -1$.

Find the general solution of $dy/dx = -y/x$, $x \neq 0$.

Solution

$$\int \frac{1}{y}\,dy = \int -\frac{1}{x}\,dx$$

$$\ln|y| = -\ln|x| + C$$

$$\ln|x| + \ln|y| = C$$

$$\ln|xy| = C \Rightarrow |xy| = e^C$$

$$y = \frac{A}{x}, \quad A \in \mathbb{R}.$$

This is a family of rectangular hyperbolas; $A = 0$ also gives a solution.

Using Reverse Product Rule

Solution 6

We will solve $y' + Py = Q$, where P, Q are functions of x. First practise reverse product rule.

Find the general solution of $x^3y' + 3x^2y = \sin x$. The variables do not separate. What do you notice about the left-hand side?

Recognise

$$x^3y' + 3x^2y = \frac{d}{dx}(x^3y).$$

Differentiate x^2y.

$$x^2y' + 2xy$$

Differentiate $y \sin x$.

$$y' \sin x + y \cos x$$

Rewrite $x^4y' + 4x^3y$.

$$\frac{d}{dx}(x^4y)$$

Rewrite $e^xy' + e^xy$.

$$\frac{d}{dx}(e^xy)$$

Rewrite $(\ln x)y' + y/x$.

$$\frac{d}{dx}(y \ln x)$$

The coefficient of y' becomes the factor multiplying y inside the derivative.

Using Reverse Product Rule

Solution 7

Solve $x^3y' + 3x^2y = \sin x$.

Solution

$$\frac{d}{dx}(x^3y) = \sin x \Rightarrow x^3y = -\cos x + C$$

$$y = \frac{C - \cos x}{x^3} \quad (x \neq 0).$$

Test: $\frac{1}{x}y' - \frac{y}{x^2} = e^x$.

$$\frac{d}{dx}\left(\frac{y}{x}\right) = e^x$$

$$\frac{y}{x} = e^x + C \Rightarrow y = x(e^x + C).$$

Test: $4xyy' + 2y^2 = x^2$.

$$\frac{d}{dx}(2xy^2) = x^2$$

$$2xy^2 = \tfrac{1}{3}x^3 + C$$

$$y^2 = \tfrac{1}{6}x^2 + \frac{K}{x} \quad (x \neq 0).$$

Solving First-order Linear Equations

Solution 8

What if reverse product rule is not immediately available? Solve

$y' - 4y = e^x$.

Multiply by the integrating factor $e^{\int P\,dx}$, where $P = -4$.

Factor

$$\text{IF} = e^{\int -4\,dx} = e^{-4x}.$$

Multiply

$$e^{-4x}y' - 4e^{-4x}y = e^{-3x}.$$

Integrate

$$\frac{d}{dx}(ye^{-4x}) = e^{-3x}$$

$$ye^{-4x} = -\tfrac{1}{3}e^{-3x} + C$$

$$y = -\tfrac{1}{3}e^x + Ce^{4x}.$$

Why the Integrating Factor Works

Solution 9

For $y' + Py = Q$, suppose $f(x)$ is an integrating factor. Multiply through:

$fy' + fPy = fQ.$

To use reverse product rule, the left side must equal $fy' + f'y$.

Comparing coefficients:

$f' = fP.$

Choose $f > 0$, divide and integrate:

$$\int \frac{f'(x)}{f(x)}\,dx = \int P\,dx$$

$$\ln f(x) = \int P\,dx.$$

$f(x) = e^{\int P\,dx}.$

A non-zero constant multiplier does not affect the method.

When the Derivative Has a Coefficient

Solution 10

Solve $\cos x\, y' + 2y \sin x = \cos^4 x$. What should we do first?

Divide

1. Divide by $\cos x$, on intervals where it is non-zero:

   $y' + 2y \tan x = \cos^3 x.$

Factor

2. Find the factor:

   $e^{\int 2\tan x\,dx} = e^{2\ln|\sec x|} = \sec^2 x.$

Multiply

3. Multiply through:

   $\sec^2 x\, y' + 2y \sec^2 x \tan x = \cos x$

   $\frac{d}{dx}(y \sec^2 x) = \cos x.$

   You can go straight to this derivative, but remember to multiply the right side by the factor too.

Integrate

4. Integrate and simplify:

   $y \sec^2 x = \sin x + C$

   $y = \cos^2 x(\sin x + C).$

Test Your Understanding — Solution 11

Edexcel FP2 (Old), June 2011 Q3

Find the general solution of

$$xy' + 5y = \frac{\ln x}{x}, \quad x > 0.$$

Factor

$$y' + \frac{5}{x}y = \frac{\ln x}{x^2}, \quad \text{IF} = e^{5\ln x} = x^5$$

$$\frac{d}{dx}(x^5 y) = x^3 \ln x.$$

Solution

Integrate by parts:

$$\int x^3 \ln x \, dx = \frac{x^4 \ln x}{4} - \int \frac{x^3}{4}\, dx$$

$$= \frac{x^4 \ln x}{4} - \frac{x^4}{16} + C.$$

$$x^5 y = \frac{x^4 \ln x}{4} - \frac{x^4}{16} + C$$

$$y = \frac{\ln x}{4x} - \frac{1}{16x} + \frac{C}{x^5}.$$

Exercise 7A — Core Pure Book 2 Pages 151–153 — Solution 12

Exercise 7A — Core Pure Book 2, pages 151–153.

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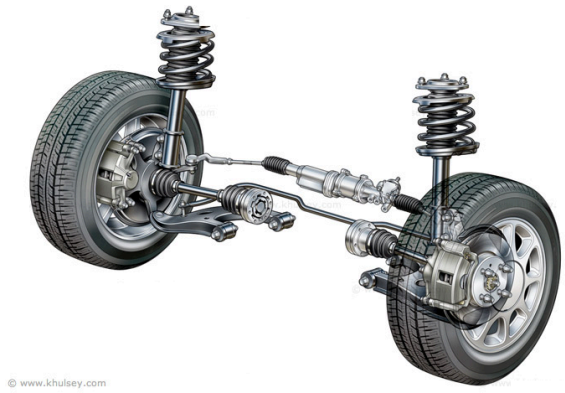
Exercise 7A — Core Pure Book 2, pages 151–153.

Second-order Differential Equations

Solution 17

Second-order differential equations involve a second derivative.

Car shock absorbers provide an example. Damping opposes velocity; the restoring spring force opposes displacement from equilibrium. Their magnitudes are proportional to velocity and extension respectively.



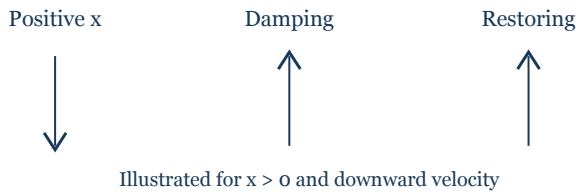
Take downward displacement x as positive, measured from equilibrium. Newton's second law gives:

$$m\frac{d^2x}{dt^2}$$

$$=$$

$$-b\frac{dx}{dt}$$

$$-cx.$$



Illustrated for x > 0 and downward velocity

For $x > 0$ and $\dot{x} > 0$, both forces and the resulting acceleration are upward. Positive coordinate direction does not specify actual acceleration.

Simple Second-order Differential Equations

Solution 18

For $ay' + by = 0$, the solution is $y = Ae^{-bx/a}$. Try a similar form for

$ay'' + by' + cy = 0, \quad a \neq 0.$

Let $y = Ae^{mx}$.

$$y' = Ame^{mx}, \qquad y'' = Am^2e^{mx}.$$

$$aAm^2e^{mx} + bAme^{mx} + cAe^{mx} = 0$$
$$Ae^{mx}(am^2 + bm + c) = 0.$$

For a non-zero trial solution, $A \neq 0$, so

$am^2 + bm + c = 0.$

This is the auxiliary equation: its roots determine the solutions of the differential equation.

Distinct Real Roots

Solution 19

If m is a root of $am^2 + bm + c = 0$, then Ae^{mx} solves $ay'' + by' + cy = 0$.

For distinct real roots α, β:

$y = Ae^{\alpha x} + Be^{\beta x}$.

Here A, B are arbitrary constants.

A zero right-hand side makes the equation homogeneous.

Solve $2y'' + 5y' + 3y = 0$.

Solution

$2m^2 + 5m + 3 = 0$

$(2m + 3)(m + 1) = 0$

$m = -\frac{3}{2},\ -1$

$y = Ae^{-3x/2} + Be^{-x}$.

Repeated Roots

Solution 20

For two equal roots α, where $b^2 - 4ac = 0$:

$y = (A + Bx)e^{\alpha x}$.

Show that $(A + Bx)e^{3x}$ satisfies

$y'' - 6y' + 9y = 0$.

The auxiliary equation is $(m - 3)^2 = 0$.

Derivatives

$y = Ae^{3x} + Bxe^{3x}$

$y' = 3Ae^{3x} + 3Bxe^{3x} + Be^{3x}$

$y'' = 9Ae^{3x} + 9Bxe^{3x} + 6Be^{3x}$.

Verify

$y'' - 6y' + 9y$

$= e^{3x}\{(9 - 18 + 9)A$

$+(9 - 18 + 9)Bx + (6 - 6)B\} = 0.$

Why not $Ae^{\alpha x} + Be^{\alpha x}$? It combines to just one arbitrary constant. The extra x gives an independent solution, analogous to the distinct terms $A/x + B/x^2$ used for repeated partial-fraction factors.

Pure Imaginary Roots

Solution 21

Complex roots work similarly to distinct real roots.
Solve $y'' + 16y = 0$.

Roots

$m^2 + 16 = 0 \Rightarrow m = \pm 4i$

$y = Ae^{4ix} + Be^{-4ix}$.

Rewrite

Using Euler's formula:

$y = A(\cos 4x + i \sin 4x)$

$+B(\cos 4x - i \sin 4x)$

$= P \cos 4x + Q \sin 4x,$

$P = A + B, \quad Q = i(A - B).$

For real solutions, take P, Q real.

For roots $\pm i\omega$:

$y = A \cos \omega x + B \sin \omega x.$

General Complex Roots

Solution 22

What about general complex roots? Solve
$y'' - 6y' + 34y = 0$.

Roots

$m^2 - 6m + 34 = 0 \Rightarrow m = 3 \pm 5i$

$y = Ae^{(3+5i)x} + Be^{(3-5i)x}$.

Rewrite

Use Euler's formula and factor out e^{3x}:

$y = e^{3x}(P \cos 5x + Q \sin 5x).$

For roots $p \pm iq$:

$y = e^{px}(A \cos qx + B \sin qx),$

with real arbitrary constants A, B.

Quickfire Questions

Solution 23

Find general solutions for these auxiliary equations and supplied roots.

$m^2 + 6m + 8 = 0$, roots $-2, -4$

$y = Ae^{-4x} + Be^{-2x}$.

$m^2 - 1 = 0$, roots ± 1

$y = Ae^{x} + Be^{-x}$.

$m^2 - 2m + 1 = 0$, roots 1

$y = (A + Bx)e^{x}$.

$m^2 + 4 = 0$, roots $\pm 2i$

$y = A\cos 2x + B\sin 2x$.

$m^2 + 10m + 25 = 0$, roots -5

$y = (A + Bx)e^{-5x}$.

$m^2 - 12m + 45 = 0$, roots $6 \pm 3i$

6

$y = e^{6x}(A\cos 3x + B\sin 3x)$.

$m^2 + 10 = 0$, roots $\pm i\sqrt{10}$

$y = A\cos(\sqrt{10}x) + B\sin(\sqrt{10}x)$.

$m^2 + 2m + 5 = 0$, roots $-1 \pm 2i$

$y = e^{-x}(A\cos 2x + B\sin 2x)$.

Exercise 7B — Core Pure Book 2 Pages 156–157

Solution 24

Exercise 7B — Core Pure Book 2, pages 156–157.

Exercise 7B — Core Pure Book 2, pages 156–157.

Exercise 7B — Core Pure Book 2, pages 156–157.

Exercise 7B — Core Pure Book 2, pages 156–157.

Exercise 7B — Core Pure Book 2, pages 156–157.

Particular Integrals

Solution 29

What if the right-hand side is a function of x?

$ay'' + by' + cy = f(x)$.

This is non-homogeneous when f is not identically zero.

First solve the homogeneous equation:

$ay'' + by' + cy = 0$.

Its general solution is the complementary function (CF).

Next find one solution of

$ay'' + by' + cy = f(x)$

by choosing an appropriate trial function and comparing coefficients. This is the particular integral (PI).

$y = \text{CF} + \text{PI}$.

The differential operator gives zero for the CF and $f(x)$ for the PI; adding gives $f(x)$.

Constant Right-hand Side

Solution 30

Find a particular integral of

$y'' - 5y' + 6y = 3$.

Try a constant $y = \lambda$.

First derivative $y' = 0$

Second derivative

$y'' = 0$

Substitute

$0 - 5(0) + 6\lambda = 3 \Rightarrow \lambda = \frac{1}{2}$.

Hence find the general solution.

General solution

For the CF:

$m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3.$

$\text{CF} = Ae^{2x} + Be^{3x}$.

Hence

$y = Ae^{2x} + Be^{3x} + \frac{1}{2}$.

Linear Right-hand Side

Solution 31

Solve $y'' - 5y' + 6y = 2x$. The forcing is linear, so try

Trial $y = \lambda x + \mu.$

First derivative $y' = \lambda$

Second derivative

$y'' = 0$

Constant Constant terms: $6\mu - 5\lambda = 0.$

Linear Coefficients of x: $6\lambda = 2.$

Solution

$\lambda = \frac{1}{3}, \quad \mu = \frac{5}{18}$

$y = Ae^{2x} + Be^{3x} + \frac{1}{3}x + \frac{5}{18}.$

Substitute:

$(6\mu - 5\lambda) + 6\lambda x = 2x.$

Quadratic Right-hand Side

Solution 32

Solve $y'' - 5y' + 6y = 3x^2$. Try a quadratic:

Trial $y = \lambda x^2 + \mu x + \nu$

First derivative

$y' = 2\lambda x + \mu$

Second derivative

$y'' = 2\lambda$

Solution

Comparing coefficients after substitution:

$6\lambda = 3, \quad 6\mu - 10\lambda = 0,$

$2\lambda - 5\mu + 6\nu = 0.$

$\lambda = \frac{1}{2}, \quad \mu = \frac{5}{6}, \quad \nu = \frac{19}{36}.$

$y = Ae^{2x} + Be^{3x} + \frac{1}{2}x^2 + \frac{5}{6}x + \frac{19}{36}.$

Exponential and Trigonometric Forcing

Solution 33

Solve $y'' - 5y' + 6y = e^x$.

Trial 1

$y = \lambda e^x$

Derivative 1

$y' = \lambda e^x$

Derivative 2

$y'' = \lambda e^x$

Solution 1

$(1 - 5 + 6)\lambda = 1 \Rightarrow \lambda = \frac{1}{2}$

$y = Ae^{2x} + Be^{3x} + \frac{1}{2}e^x.$

Solve $y'' - 5y' + 6y = 13\sin 3x$.

Trial 2

$y = \lambda \sin 3x + \mu \cos 3x$

Derivative 3

$y' = 3\lambda \cos 3x - 3\mu \sin 3x$

Derivative 4

$y'' = -9\lambda \sin 3x - 9\mu \cos 3x$

Solution 2

$-3\lambda + 15\mu = 13, \quad -15\lambda - 3\mu = 0$

$\lambda = -\frac{1}{6}, \quad \mu = \frac{5}{6}$

$y = Ae^{2x} + Be^{3x} - \frac{1}{6}\sin 3x + \frac{5}{6}\cos 3x.$

When the Trial Appears in the CF

Solution 34

A trial PI cannot already belong to the CF. This is like needing independent solutions for repeated auxiliary roots.

Solve $y'' - 5y' + 6y = e^{2x}$.

The CF is $Ae^{2x} + Be^{3x}$. What goes wrong if we try $y = \lambda e^{2x}$?

Failure

Then

$y = (A + \lambda)e^{2x} + Be^{3x}.$

This is still just the CF: substitution gives $0 = e^{2x}$, impossible.

Instead try

New trial $y = \lambda x e^{2x}.$

Coefficient

$y' = \lambda(1 + 2x)e^{2x}$

$y'' = \lambda(4 + 4x)e^{2x}$

$-\lambda e^{2x} = e^{2x} \Rightarrow \lambda = -1.$

Solution

$y = Ae^{2x} + Be^{3x} - xe^{2x}.$

A Constant Term in the CF

Solution 35

The PI must be independent of the CF. Solve $y'' - 2y' = 3$.

Roots $m^2 - 2m = 0 \Rightarrow m = 0, 2.$

CF

$$\text{CF} = Ae^{0x} + Be^{2x} = A + Be^{2x}.$$

A constant trial is already in the CF. Instead use

PI

$$y = \lambda x.$$

Solution

$-2\lambda = 3 \Rightarrow \lambda = -\frac{3}{2}$

$y = A + Be^{2x} - \frac{3}{2}x.$

Multiply a duplicated trial by x until it is independent of the CF. A repeated root may require x^2, rather than only one extra x.

Summary So Far

Solution 36

Right-hand side	Trial particular integral
k	λ
$ax + b$	$\lambda + \mu x$
$ax^2 + bx + c$	$\lambda + \mu x + \nu x^2$
ke^{px}	λe^{px}
$m\cos\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$
$m\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$
$m\cos\omega x + n\sin\omega x$	$\lambda\cos\omega x + \mu\sin\omega x$

Modify a trial which duplicates a CF term by multiplying by a sufficient power of x. An exam question may suggest the modified trial.

Test Your Understanding — Solution 37

June 2010 Q8

(a) Find λ for which $y = \lambda x \sin 5x$ is a PI of $y'' + 25y = 3\cos 5x$.

(4 marks)

Notice the extra factor x.

a

$y' = \lambda \sin 5x + 5\lambda x \cos 5x$

$y'' = 10\lambda \cos 5x - 25\lambda x \sin 5x$

$y'' + 25y = 10\lambda \cos 5x$

$\lambda = \frac{3}{10}$.

(b) Hence find the general solution. (3 marks)

b

$m^2 + 25 = 0 \Rightarrow m = \pm 5i$

$\text{CF} = A\cos 5x + B\sin 5x$

$y = A\cos 5x + B\sin 5x + \frac{3}{10}x\sin 5x.$

Test Your Understanding — Solution 38

June 2012 Q4

Find the general solution of

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 2\cos t - \sin t.$$

Here x is the dependent variable and t the independent variable.

Solution

$m^2 + 5m + 6 = 0 \Rightarrow m = -2, -3$

$\text{CF} = Ae^{-2t} + Be^{-3t}$.

Try $x = P\cos t + Q\sin t$:

$\dot{x} = -P\sin t + Q\cos t, \quad \ddot{x} = -P\cos t - Q\sin t.$

$(-P\cos t - Q\sin t) + 5(-P\sin t + Q\cos t)$

$+6(P\cos t + Q\sin t) = 2\cos t - \sin t.$

$-P + 5Q + 6P = 2, \quad -Q - 5P + 6Q = -1$

$5P + 5Q = 2, \quad -5P + 5Q = -1$

$P = \frac{3}{10}, \quad Q = \frac{1}{10}.$

$x = Ae^{-2t} + Be^{-3t} + \frac{3}{10}\cos t + \frac{1}{10}\sin t.$

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2, pages 161–162.

Exercise 7C — Core Pure Book 2 Pages 161–162 Solution 43

Exercise 7C — Core Pure Book 2, pages 161–162.

Boundary Conditions Solution 44

Conditions let us determine arbitrary constants. Find y given

$y'' - y = 2e^x, \qquad y(0) = y'(0) = 0.$

General solution $y = Ae^x + Be^{-x} + xe^x.$

Use y(0)

$0 = Ae^0 + Be^0 + 0 \Rightarrow A + B = 0.$

Differentiate

$y' = Ae^x - Be^{-x} + xe^x + e^x.$

Use derivative $0 = A - B + 1 \Rightarrow A - B = -1.$

Constants $A = -\frac{1}{2}, \quad B = \frac{1}{2}.$

Solution

$y = -\frac{1}{2}e^x + \frac{1}{2}e^{-x} + xe^x.$

Test Your Understanding

Solution 45

June 2010 Q8 revisited

$y'' + 25y = 3\cos 5x$.

(c) Given $y(0) = 0$, $y'(0) = 5$, find $y = f(x)$. (5 marks)

Earlier:

$y = A\cos 5x + B\sin 5x + \frac{3}{10}x\sin 5x$.

c

At $x = 0$, $y = 0 \Rightarrow A = 0$.

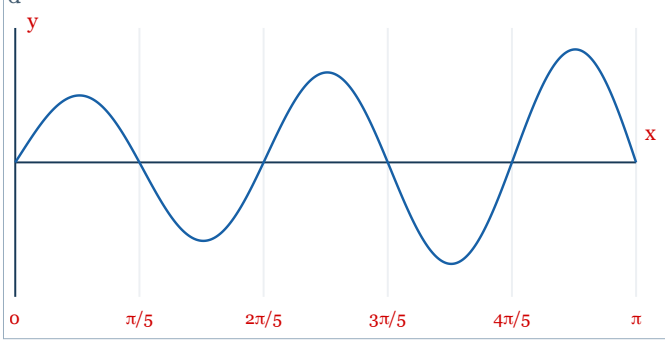
$y' = 5B\cos 5x + \frac{3}{10}\sin 5x + \frac{3}{2}x\cos 5x$

$5 = 5B \Rightarrow B = 1$.

$y = (1 + \frac{3}{10}x)\sin 5x$.

(d) Sketch $y = f(x)$ for $0 \le x \le \pi$. (2 marks)

d



Features

The curve passes through the origin and oscillates with increasing amplitude. Its zeros are $x = 0, \pi/5, 2\pi/5, 3\pi/5, 4\pi/5, \pi$.

Exercise 7D — Core Pure Book 2 Pages 164–165

Solution 46

Exercise 7D — Core Pure Book 2, pages 164–165.

Exercise 7D — Core Pure Book 2, pages 164–165.

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